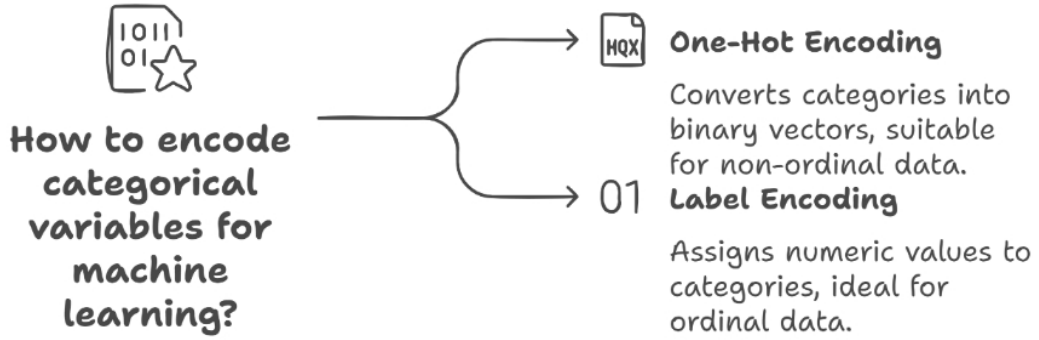
Machine learning models typically require numeric inputs. Categorical variables, such as labels or categories, must be converted into numeric formats. Scikit-Learn provides robust tools for encoding categorical variables, ensuring they are represented effectively in the dataset.



One-Hot Encoding

One-hot encoding transforms categorical variables into binary vectors. Each category is represented as a separate column with binary values indicating its presence.

```
from sklearn.preprocessing import OneHotEncoder

# Initialize the encoder
encoder = OneHotEncoder()
data_encoded = encoder.fit_transform(data[['categorical_column']]) Alternatively, Pandas' get_dummies() method provides a simple way to achieve one-hot encoding: import pandas as pd

data_encoded = pd.get_dummies(data, columns=['categorical_column'])
```

Label Encoding

For ordinal data or when categories have a natural order, label encoding assigns a unique integer to each category: from sklearn.preprocessing import LabelEncoder

```
encoder = LabelEncoder()
data['encoded_column'] = encoder.fit_transform(data['categorical_column'])
```

Feature Scaling and Normalization

Scaling ensures that features have comparable ranges, which is critical for algorithms sensitive to feature magnitudes, such as gradient descent-based models.